

UX / UI CASE STUDY — MOBILE APPLICATION — FIGMA 2024

FixSense

A complete UX/UI design case study for a mobile application that simplifies car diagnostics, service booking, and maintenance tracking for everyday drivers.

PROJECT TYPE	DURATION	ROLE	TOOL
Mobile Application	6 Weeks	UX/UI Designer	Figma

Project Overview

FixSense is a mobile application designed to simplify the car maintenance experience. The app allows users to find trusted repair shops, explore available services, book appointments, and manage vehicle maintenance — all in one place. The goal is to provide a simple, reliable, and user-friendly solution for every car owner.

6wks

PROJECT DURATION

5

USER PERSONAS

30+

SCREENS DESIGNED

5

KEY USER FLOWS

Problem Statement

Car owners often struggle to find reliable repair shops, compare available services, and book maintenance appointments efficiently. Many existing solutions are either difficult to use or provide limited information, making the maintenance process stressful and time-consuming.

Can't diagnose

Users have no way to understand what's wrong with their car.

Hard to find shops

No reliable way to discover and compare nearby repair centers.

Complex booking

Booking an appointment requires multiple phone calls.

No transparency

Users fear being overcharged with no price visibility.

Project Goals

- Simplify the maintenance process for all types of car owners.
- Help users find and compare trusted repair shops nearby.
- Make service booking fast, clear, and stress-free.
- Save users time through intelligent diagnostics.
- Provide a clean, intuitive, and accessible user experience.

Competitive Analysis

Six existing car maintenance apps were analysed to identify gaps and opportunities that FixSense could address.

APP	KEY STRENGTHS	KEY WEAKNESSES
Carly	One-click coding, odometer fraud detection, DIY guidance	Expensive subscriptions, requires proprietary hardware
OBDeleven	Full system diagnostics, hybrid battery monitoring	High hardware cost, complex UI for beginners
FIXD	Simple engine code translation, multi-vehicle support	Many features behind monthly fees, no advanced coding
Drivvo	Fleet expense management, fuel efficiency charts	Tedious manual data entry, cluttered UI
Carista	One-click service resets, clean interface	Useless without Carista hardware, no service reminders
Simply Auto	Multi fuel type support, cloud backup	Manual entry heavy, outdated UI design

SWOT Analysis

Strengths

Fast fault detection · Saves time for drivers · Shows nearby service centers · Easy to use

Weaknesses

Requires internet connection · Some faults may be misdiagnosed · Not all service centers included

Opportunities

Increasing number of cars · Growing reliance on apps · Possibility to add online booking

Threats

Competing apps · Changing repair prices · Users may not trust online diagnosis

User Personas

Five distinct user personas were developed through research to represent the key target audiences for FixSense.

AH

Ahmed Hassan

Sales Representative

29 · Cairo, Egypt

GOALS

Feel confident when driving, understand car issues without technical knowledge, get help quickly when something goes wrong.

FRUSTRATIONS

Does not understand warning lights. Mechanics use complex terms. Fear of being overcharged. Difficulty finding a reliable mechanic quickly.

SK

Sara Khaled

English Teacher

28 · Tanta, Gharbia, Egypt

GOALS

Quickly understand what is wrong with the car, know whether the problem is serious, feel safe while driving, find a nearby trusted repair center.

FRUSTRATIONS

Feels worried and scared when a warning light appears, especially when driving alone. Usually calls family or searches online.

SM

Saber Mahmoud

Taxi Driver

35 · Shibin El-Kom, Monoufia

GOALS

Keep car in good condition, quickly understand problems, find a nearby repair center to minimize downtime and get back to work fast.

FRUSTRATIONS

Any car problem directly affects his income. Drives long hours daily and needs fast, reliable solutions.

ME

Mona El-Sayed

University Student

21 · Alexandria, Egypt

GOALS

Drives a hand-me-down family car and wants to learn basic maintenance, avoid being overcharged as a young woman at the mechanic, and book services without phone calls.

FRUSTRATIONS

Feels mechanics don't take her seriously. Limited budget for repairs. Doesn't know which maintenance is genuinely necessary versus upsold.

User Personas

KF

Karim Farouk

Small Business Owner, Delivery Fleet

41 · Giza, Egypt

GOALS

Manages three delivery vehicles and needs to track maintenance schedules across the fleet, minimize unexpected breakdowns, and keep cost records for the business.

FRUSTRATIONS

Juggling multiple cars' service histories manually. No single place to compare workshop pricing across vehicles. Downtime directly cuts into delivery revenue.

06 — Empathy Map

Key insights about what our primary user says, thinks, does, and feels when experiencing a car problem.

Says

"The car broke down suddenly." · "I don't understand the problem." · "I need a mechanic quickly." · "How much will it cost?"

Thinks

"Is the problem serious or minor?" · "Is the mechanic reliable?"

Does

Searches on Google · Opens Maps · Calls a friend · Waits for help

Feels

Nervous · Worried · In a hurry · Confused

Pains

Doesn't know how to diagnose the problem · Afraid of being scammed

Gains

Find out the problem quickly · Find a nearby mechanic · Know an approximate cost

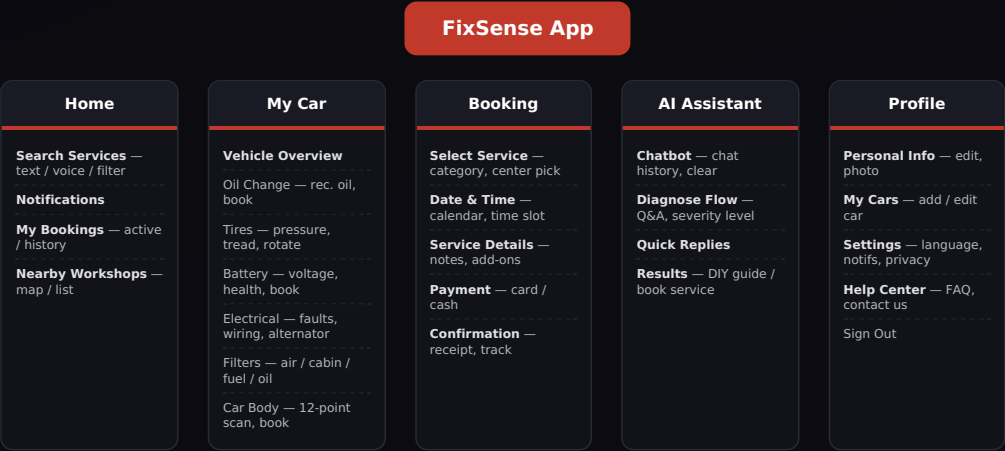
User Journey Map

Mapping the complete experience of a user from the moment their car breaks down to leaving a review after service.

#	ACTION	GOAL	FEELING	PAIN POINT	OPPORTUNITY
1	Car breaks down	Understand problem	Worried & stressed	Doesn't know cause	Quick diagnosis
2	Opens the app	Start solving issue	Unsure	Doesn't know where to start	Simple onboarding
3	Enters Diagnose	Identify the issue	Slightly relieved	Might pick wrong option	Clear UI with icons
4	Answers questions	Get accurate diagnosis	Focused	Questions may confuse	Guided questions
5	Views result	Understand the issue	Curious	Not fully confident	Show accuracy %
6	Checks solution	Decide next step	Comfortable	Can't fix alone	DIY + service option
7	Chooses shop	Fix the car	Reassured	Afraid of high cost	Ratings + price range
8	Books appointment	Save time	Relaxed	Booking may be complex	Easy booking flow
9	Leaves review	Share experience	Satisfied	Might skip feedback	Quick review UI

Information Architecture

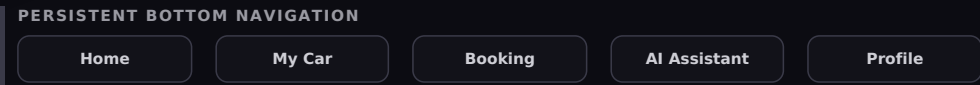
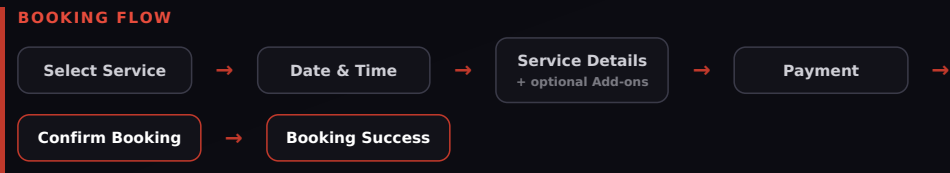
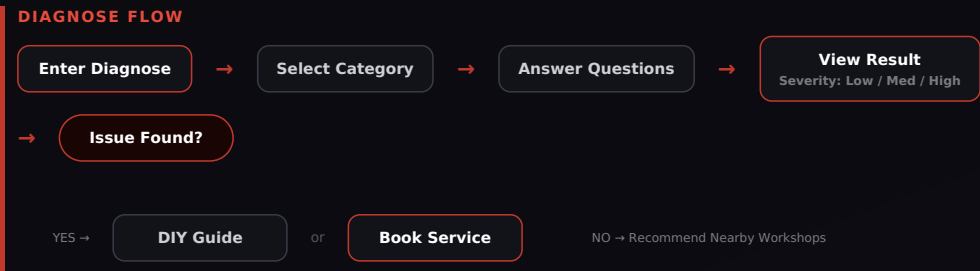
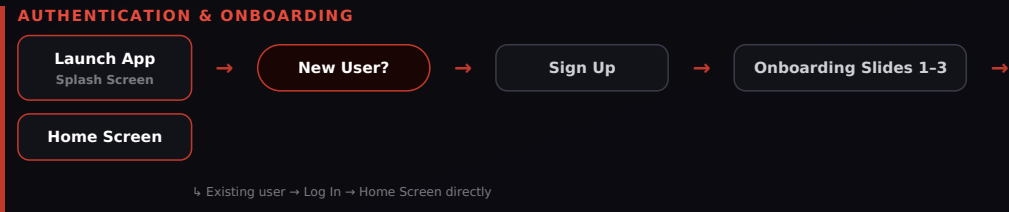
The app is structured around five main navigation tabs, each housing a set of screens and sub-flows to cover the complete car care journey.



Level 0 — Root App Level 1 — Main Tabs Level 2/3 — Screens & Sub-screens

Primary User Flow

Onboarding → Diagnose → Book Service. The core flow connecting authentication, diagnosis, and the booking journey across the app.



Key Screens & Features

01	SPLASH & ONBOARDING	Clean welcome screen with app tagline "Everything you need fixed in one place." Three onboarding slides introduce core features before the Sign In screen.
02	HOME SCREEN	Personalized greeting, vehicle health status card, quick service shortcuts (Oil, Brakes, Tires, Battery), upcoming booking reminder, and nearby workshops.
03	MY CAR — VEHICLE HEALTH	Full diagnostic overview with system health percentages for Tires, Battery, Oil, Electrical, Filters, and Car Body. Each system links to detailed view and nearest workshops.
04	BOOKING FLOW (4 STEPS)	Step 1: Select service category. Step 2: Pick date and time slot. Step 3: Review details and optional add-ons. Step 4: Confirm with payment method.
05	AI ASSISTANT / CHATBOT	Conversational interface for guided car diagnosis. User can type or use quick reply chips. Suggests service centers at the end of the diagnosis flow.
06	USER PROFILE	Manage personal info, registered vehicles, app settings, language preference, help center, and contact support. Includes sign-out option.

Visual Design & Style

Color Palette

- Background Dark #0a0a0f — Primary background
- Background Mid #0d0d14 — Cards, nav bars
- Primary Red #c0392b — CTAs, highlights
- Red Tint #1a0505 — Icon backgrounds
- White #ffffff — Primary text
- Grey #888888 — Secondary text

Typography & Components

ELEMENT	FONT	SIZE
Heading 1	Inter Bold	28-32px
Heading 2	Inter SemiBold	18-22px
Body Text	Inter Regular	14-16px
Caption	Inter Regular	12px
Button	Inter Bold	14px

Challenges & Solutions

- Grid & Spacing Consistency**

Problem: Early screens had inconsistent spacing and element alignment across different screen sizes.

Solution: Implemented a strict 8px grid system and 4-column layout in Figma with auto-layout components.
- User Flow Logic**

Problem: The Diagnose feature was disconnected from the Service Centers flow, leaving users stranded.

Solution: Redesigned the flow so that diagnose results always surface nearby service center recommendations.
- Information Overload**

Problem: The My Car screen displayed too much technical data, overwhelming non-technical users.

Solution: Simplified to health percentage cards with color indicators, hiding raw data behind a "details" tap.
- Booking Complexity**

Problem: Initial booking flow had 7 steps which user testing showed was too long.

Solution: Condensed to a 4-step flow with optional add-ons collapsed by default.

Key Outcomes & Learnings

✓	Designed 30+ screens covering the complete user journey from onboarding to booking.
✓	Established a consistent dark-theme design system reusable across all screens.
✓	Resolved professor-flagged UX issues: connected Diagnose flow to Service Centers.
✓	Reduced booking steps from 7 to 4 while maintaining all necessary information.
✓	Created five detailed user personas grounded in real Egyptian car owner behavior.
✓	Delivered a high-fidelity prototype in Figma ready for developer handoff.

Key Learnings

USER-FIRST FLOWS	Every screen transition must serve a clear user need. Disconnected flows break trust.
SIMPLICITY WINS	Non-technical users need layered information — summary first, details on demand.
CONSISTENCY MATTERS	A strict design system (grid, spacing, color) saves iteration time and looks professional.
ITERATE FAST	Professor feedback revealed blind spots invisible after hours of solo design work.

Thank You

This case study documents the full UX/UI design process for FixSense — from user research and competitive analysis through information architecture, user flows, wireframing, and high-fidelity Figma screens.

Tool: Figma | Duration: 6 Weeks | Role: UX/UI Designer